

Vertical Pod Autoscaler (VPA)

Since you just learned **HPA**, VPA is the other important autoscaling concept.

1. What is VPA?

VPA = Vertical Pod Autoscaler

VPA automatically adjusts the CPU and memory resources requested by Pods according to their resource usage.

Simple example:

A Pod currently has:

CPU: 500m

Memory: 512Mi

If the application needs more resources, VPA may recommend or apply higher values, depending on its configuration:

CPU: 1000m

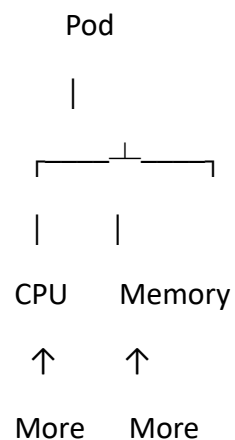
Memory: 1Gi

Easy memory:

VPA = Makes the Pod "bigger" or "smaller" in terms of CPU and memory resources.

2. Why is it called "Vertical"?

Because **vertical scaling means increasing or decreasing the resources given to a Pod.**



resources resources

You are not adding more Pods.

Instead, you're adjusting the resources of the existing Pod.

3. HPA vs VPA ★★ ★

This is **very important for KCNA**.

HPA

Adds or removes Pods:

2 Pods



5 Pods

VPA

Changes CPU/memory resources:

Pod



More CPU

More Memory

🧠 Easy trick:

HPA → How many Pods?

VPA → How much CPU/RAM?

4. Real-Life Example 🏠

Imagine you have a small room.

HPA:

Instead of making the room bigger, you add more rooms:



More Pods

VPA:

You make the existing room bigger:

Small Room



Large Room

More CPU/Memory for the Pod

5. What does VPA adjust?

VPA mainly works with:

CPU

Example:

500m → 1000m

Memory

Example:

512Mi → 1Gi

These values are associated with the container's **resource requests and limits**, depending on the VPA configuration and mode.

6. Resource Requests and VPA

Remember your earlier **Resource Limits** topic.

A container can have:

resources:

requests:

cpu: "500m"

memory: "512Mi"

The request tells Kubernetes approximately:

"I need this amount of CPU and memory."

VPA observes the workload's resource usage and can provide recommendations or modify resource requests depending on how it is configured.

7. VPA Modes ★

VPA has different operating modes.

1 Off

VPA **only provides recommendations.**

VPA



Recommendation



You decide what to change

It does not automatically change the Pod resources.

2 Auto

VPA can automatically apply recommended resource values.

Depending on the required change, this may involve restarting/recreating Pods so that the new resource settings can take effect.

VPA



New recommendation



Pod updated/recreated



New resources

3 Recreate

VPA updates resources by **recreating Pods** when necessary.

Old Pod



Recreate



New Pod



New resources

4 Initial

VPA applies recommendations **when Pods are initially created**.

It does not continuously update already-running Pods.

Create Pod



VPA recommendation



Pod starts with recommended resources

8. Important: VPA May Restart Pods ⚠

This is an important concept.

If VPA needs to change the resource requests of an already-running Pod, the Pod may need to be **recreated**.

For example:

Running Pod

CPU = 500m



VPA recommends 1000m



Pod may be recreated



New Pod

CPU = 1000m

So VPA isn't simply "adding CPU to a running container instantly."

9. Does VPA Add More Pods?

✗ No.

VPA does not primarily increase the number of replicas.

Example:

Before:

Pod 1

Pod 2

VPA



Pod 1 → More CPU/Memory

Pod 2 → More CPU/Memory

The number of Pods remains the same.

For adding/removing Pods, you use **HPA**.

10. Does VPA Add Nodes?

✗ No.

VPA adjusts/recommends resources for Pods.

If the cluster doesn't have enough Node capacity, another mechanism such as **Cluster Autoscaler** may be needed.

VPA



Pod needs more resources



Not enough Node capacity



Cluster Autoscaler



May add Node

11. VPA Components

VPA commonly has three important components:

◆ VPA Recommender

Analyzes resource usage and recommends appropriate CPU/memory values.

Usage data



Recommender



Recommendation

◆ VPA Updater

Checks running Pods and determines which Pods need to be updated based on VPA recommendations.

◆ VPA Admission Controller

Applies recommended resource values when a Pod is created.

Easy memory:

Recommender



"How much resource should it have?"

Updater



"Which running Pod needs updating?"

Admission Controller



"Apply resources when Pod is admitted."

12. VPA Example

Imagine your application has:

Pod 1

CPU request: 100m

Memory request: 128Mi

But over time, the application regularly needs:

CPU: 500m

Memory: 512Mi

VPA can analyze this usage and recommend:

CPU: 500m

Memory: 512Mi

Depending on the VPA mode, those recommendations may simply be shown or automatically applied.

13. VPA vs HPA vs Cluster Autoscaler ★★

Type	What does it change?	Easy meaning
HPA	Number of Pods	⊕/⊖ Pods
VPA	CPU/Memory resources	⬆/⬇ Pod resources
Cluster Autoscaler	Number of Nodes	⊕/⊖ Nodes

🧠 Remember:

HPA → How many Pods? 📦📦📦

VPA → How much CPU/RAM? 💻

Cluster Autoscaler → How many Nodes? 🖥️🖥️

★ KCNA Revision Notes

Vertical Pod Autoscaler

- **VPA = Vertical Pod Autoscaler**
- Adjusts/recommends **CPU and memory resources** for Pods.
- It is about **vertical scaling**.
- It does **not** primarily add or remove Pods.
- It can analyze historical/current resource usage to make recommendations.
- VPA can operate in different modes such as **Off, Initial, Recreate, and Auto**.

- Changing resources of an existing Pod may require the Pod to be recreated.
- **Recommender** calculates recommendations.
- **Updater** helps update Pods that need changes.
- **Admission Controller** applies recommendations when Pods are admitted.

★ **One-line definition:**

VPA automatically manages or recommends CPU and memory resource requests for Pods based on their resource usage.

🧠 **Final memory trick:**

HPA → More/Fewer Pods 📦📦📦

VPA → More/Less CPU & Memory 💻

CA → More/Fewer Nodes 🖥️🖥️